INCH-POUND

GGG-B-440C
10 July 2026
SUPERSEDING
GGG-B-440B
25 September 1964

FEDERAL SPECIFICATION

BLADE, CIRCULAR SAW, WOODCUTTING

Inactive for new design after 29 July 2008

The General Services Administration has authorized the use of this federal specification by all federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers circular saw blades for bench-mounted and floor-mounted, woodcutting circular saws, power-driven saws, and portable circular saws (see 6.1).

1.2 Classification. Circular saw blades shall be of the following types and classes as specified (see 6.2).

Type I - Crosscut.
  Class 1 - Flat ground.
  Class 2 - Hollow ground.
Type II - Rip.
  Class 1 - Flat ground.
Type III - Combination.
  Class 1 - Flat ground.
  Class 2 - Hallow ground.
Type IV - Dado heads.

2. APPLICABLE DOCUMENTS

2.1 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://www.asq.org.)

Comments, suggestions, or questions on this document should be addressed to DLA Troop Support – Construction and Equipment (C&E) (ATTN: Code FCSB), 700 Robbins Avenue, Philadelphia, PA 19111-5098 or email to CE_Standardization@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 3230

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-E3024/E3024M - Standard Practice for Magnetic Particle Testing for General Industry

(Copies of this document are available from https://www.astm.org.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Circular saw blades shall be made of steel having essentially the chemical and mechanical properties shown in Tables I and II and shall be free from cracks, segregations, seams, hammer marks, high spots or other imperfections that may affect their serviceability.

3.1.1 The values designated in Tables I and II are nominal and not restrictive. Comparable blade material may be used provided the impact strength, cutting wear resistance, and thermal stability properties equal or exceed those afforded by the specified steel.

TABLE I. Chemical composition.

Carbon	Manganese (maximum)	Nickel (minimum)	Silicon	Chromium (maximum)	Phosphorus (maximum)	Sulphur (maximum)	Molybdenum (maximum)
Percent 0.70-0.80	Percent 0.70	Percent 0.60	Percent 0.20-0.35	Percent 0.25	Percent 0.025	Percent 0.025	Percent 0.20

TABLE II. Mechanical properties (nominal) at Rockwell C45.

Tensile strength	215,000 psi
Yield strength	195,000 psi
Elongation at 2 inches (percent)	6
Reduction area (percent)	25

3.2 Uniformity of thickness. The nominal thickness of the blades is shown in tables III through VII, inclusive, (for the tolerance, see 3.6). The nominal thickness of the flat ground blades shall be designated by a single dimension covering every part of the blade. Uniformity of thickness for both flat ground and hollow ground blades, along any circumference generated by a radius, shall be uniform within plus or minus 0.001 inch. For flat ground only, the thickness in any radial direction shall be uniform within plus or minus 0.0015 inch. Hollow ground blades, which are thinner in back of the rim for clearance, are designated by three dimensions, at the rim, the thinnest circle, and the collar, in that sequence.

3.3 Hardness. All blades, except hollow ground, shall be hardened and tempered to a Rockwell C hardness of not less than 42 nor more than 48. Hollow ground blades shall have a Rockwell C hardness of not less than 46 nor more than 62.

3.4 Tooth profile. The profile of teeth shall conform to the styles shown on Figure 1 and as specified herein for the individual types and classes.

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STYLE-A



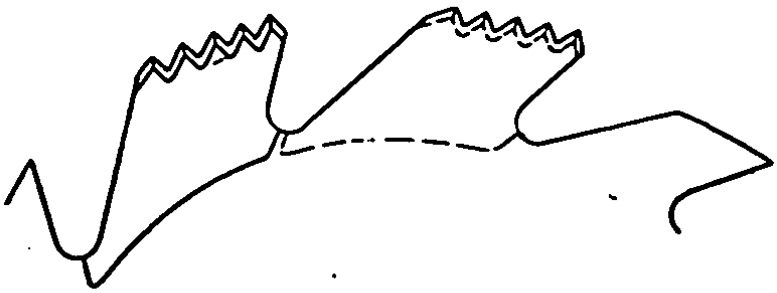
STYLE-U



STYLE-V



STYLE-L



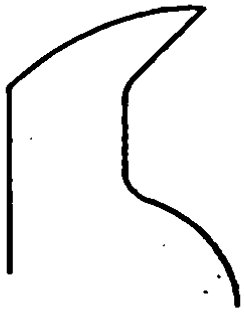
STYLE-Y



STYLE-M



STYLE-P



STYLE-BB



STYLE-S



STYLE-SPECIAL

Figure 1. Saw blade teeth profiles.

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3.5 Arbor holes. Unless otherwise specified (see 6.2), the arbor holes shall be round. The arbor hole diameter for all blades shall be as specified (see 6.2). If specified, blades up to and including 12 inches in diameter with round arbor holes shall be furnished with adapter bushings as follows:

6- through 8-inch blades---0.625-inch arbor hole bushed to 0.500 inch.
9- through 10-inch blades---0.750-inch arbor hole bushed to 0.625 inch.
12-inch blades---1-inch arbor hole bushed to 0.750 inch.

3.6 Tolerances. The following tolerances will be permitted:

Arbor holes:
Plus 0.003 inch and no minus tolerances
Nominal rim diameters:
6- to 10-inch diameter blades, plus 0.063 inch, minus 0.
10- to 24-inch diameter blades, plus 0.125 inch, minus 0
Over 24-inch diameter blades, plus 0.187 inch, minus 0
Nominal blade thickness:
Flat and hollow ground blades, 10-inch diameter and under, plus or minus 10 percent; over 10-inch diameter, plus or minus 6 percent
Number of teeth:
Plus or minus 10 percent
Lateral runout:
Lateral or transverse runout as measured just inside the tooth gullet line shall not exceed 0.0004-inch of blade diameter---static measurement total indicating reading (T.I.R).
Peripheral runout:
Peripheral runout as measured at the tips of the cutting teeth shall not exceed 0.0006-inch per inch of blade diameter---static measurement T.I.R.

3.7 Identification marking. Each blade shall be permanently marked with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined. The marking shall be applied in a manner that will preclude damage to the blades.

3.8 Finish. The sides of all blades shall be uniformly ground to a bright, smooth finish of 63 microinches or less. When specified (see 6.2), chrome plated blades shall be furnished.

3.9 Tensioning. Blades up to and including 10-inch diameter blades shall be tensioned for 7000 revolutions per minute (RPM). Blades 12 inches or over in diameter shall be tensioned for a rim speed of 10,000 surface feet per minute (SFM). When specified (see 6.2), blades 12, 14, and 16-inch diameter sizes shall be tensioned for 3000 revolutions per minute (RPM).

3.10 Type I, crosscut.

3.10.1 Class 1, flat ground. The blades shall have style L teeth, as shown on figure 1, shall be spring set and bevel filed, and shall be adapted for crosscutting. The blades shall conform to the dimensions and number of of teeth shown in table III, as specified (see 6.2). When specified (see 6.2), blades 30 inches and larger in diameter shall have style M teeth as shown on figure 1.

3.10.2 Class 2, hollow ground. The blades shall have style P teeth, as shown on figure 1, shall be bevel filled, and shall be adapted for crosscutting. The blades shall be hollow ground and conform to the dimensions and number of teeth shown in table IV, as specified (see 6.2).

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TABLE III. Type I, class 1, flat ground, crosscut blades.

Diameter (Inches)	Thickness (Inch)	Number of teeth
6	0.058	100
7	.058	100
8	.058	100
10	.065	100, 150
12	.083	70, 100, 150
12	.109	70, 100
14	.083	60, 80, 100, 150
14	.095	80, 100
16	.083	60, 80, 100, 150, 200
16	.095	60, 80, 100, 150
16	.109	60, 80, 100
18	.095	60, 80, 100
18	.109	60, 80, 100
18	.134	80
20	.095	80
20	.109	60, 70
20	.134	70
22	.109	70
22	.134	60
24	.120	70
24	.134	70
26	.134	70
28	.134	70
30	.134	70
30	.148	70
42	.165	80

TABLE IV. Type I, class 2, hollow ground, crosscut blades.

Diameter (Inches)	Thickness (see 3.2)			Number of teeth
	Rim (Inch)	Thinnest circle (Inch)	Collar (Inch)	
8	0.95	0.058	0.095	100
10	.109	.065	.109	100
12	.120	.072	.120	120, 150
14	.134	.083	.134	140
16	.134	.083	.134	130
18	.148	.095	.148	150
18	.165	.109	.165	100, 150
20	.180	.120	.180	100, 150
22	.180	.120	.180	100, 140
24	.203	.134	.203	120
24	.203	.120	.203	120

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3.11 Type II, rip.

3.11.1 Class 1, flat ground. The blades shall have style A teeth, as shown on figure 1, shall be filed and spring set, and shall be adapted for ripping. The blades shall conform to the dimensions and number of teeth as shown in table V, as specified (see 6.2).

TABLE V. Type II, class 1, flat ground, rip blades.

Diameter (Inches)	Thickness (Inch)	Number of teeth
6	0.058	40
7	.058	40
8	.058	36
10	.065	36
10	.083	30, 36
10	.095	30
10	.109	24, 30
12	.083	24, 30, 36
12	.109	24, 30, 36
14	.083	36
14	.109	36
16	.083	36
16	.095	36
16	.109	36
18	.095	30, 36
18	.109	30, 36
20	.095	36
20	.109	36

3.12 Type III, combination.

3.12.1 Class 1, flat ground. The blades shall have teeth conforming to style S, U, or V, or style Special, as shown on figure 1, and as specified (see 6.2). The teeth shall be filed and set in accordance with the regular trade practice and adapted for crosscutting, ripping and mitering. The blades shall conform to the dimensions and number of teeth ot teeth sections as shown in table VI, as specified (see 6.2).

3.12.2 Class 2, hollow ground. The blades shall have style S teeth as shown in figure 1. The teeth shall be filed in accordance with the regular trade practice. The blades shall be hollow ground and conform to the dimensions and number of teeth sections as shown in table VII, as specified (see 6.2).

3.13 Type IV, dado heads. Dado heads shall consist of two hollow-ground outside blades with one or more filler blades of any of the thickness shown in table VIII, as specified (see 6.2). The outside blades shall be hollow ground and of uniform thickness at rim and collar. The outside shall have style Y teeth, and the filler blades shall have style BB teeth, as shown on figure 1. The two sets of cutting teeth shall be beveled alternately to the right and left. The blades shall conform to the dimensions as shown in table VIII, as specified (see 6.2). The diameter of the blades shall be within a tolerance of plus 0.005 inch, minus 0.003 inch.

3.14 Workmanship. The workmanship, including finish and appearance, shall be first-class and equal in every respect to the best commercial practice.

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3.15 Illustrations. The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of saw blades which are otherwise in accordance with the requirements of the specification.

TABLE VI. Type III, class 1, flat ground, combination blades.

Diameter (Inches)	Thickness (Inch)	Number of teeth or sections	Style of teeth
		Number of sections	
6	0.058	10, 12	S
7	.058	10, 14	S
8	.058	12, 14	S
9	.065	12, 14, 16	S
10	.065	16	S
12	.083	18	S
14	.095	18, 20	S
16	.095	20, 22	S
8	.058	16	U
9	.065	16	U
10	.065	18	U
12	.083	20	U
14	.083	22	U
16	.095	24, 26	U
20	.109	28	U
		Number of teeth	
6	.058	20	V
7	.058	20 – 24	V
8	.058	20 – 24	V
9	.065	20 – 30	V
10	.065	44	V
12	.083	44	V
14	.083	44	V
14	.095	44	V
16	.095	44	V
18	.109	44	V
		Number of sections	
6	.058	8, 12	Special
7	.058	8, 12	Special
8	.065	8, 12	Special
9	.065	8, 12	Special
10	.083	8, 12	Special
12	.083	8, 12	Special
14	.083	8, 12	Special
16	.109	8, 12	Special

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TABLE VII. Type III, class 2, hollow ground, combination blades.

Diameter (Inches)	Thickness (see 3.2)			Number of sections
	Rim (Inch)	Thinnest circle (Inch)	Collar (Inch)	
6	0.065	0.042	0.065	10, 12
7	.083	.058	.083	10, 14
8	.083	.058	.083	12, 14
9	.083	.058	.083	12, 14, 16
10	.095	.065	.095	16
12	.109	.072	.109	18
14	.120	.083	.120	18, 20
16	.120	.083	.120	20, 22
18	.134	.095	.134	22, 24
20	.134	.095	.134	24, 26
24	.148	.109	.148	30

TABLE VIII. Type IV, dado heads.

Diameter of outside blades and fillers (Inches)	Thickness of outside blades (Inch)	Thickness of filler blades (Inch)
5	0.125	0.063, 0.125, 0.250
6	.125	.063, .125, .250
7	.125	.063, .125, .250
8	.125	.063, .125, .250
9	.125	.063, .125, .250
10	.125	.063, .125, .250
11	.125	.063, .125, .250
12	.125	.063, .125, .250
14	.125	.063, .125, .250
16	.125	.063, .125, .250
18	.125	.063, .125, .250

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his/her own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the documents where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance inspection.

4.2.1 Lot. All blades of the same type, class, diameter, and style of teeth offered for delivery at one time shall be considered a lot.

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4.2.2 Sampling for visual and dimensional examination. A random sample of blades shall be selected from each inspection lot of material with lot acceptance based on the single sampling inspection requirements in accordance with ASQ Z1.4 and table IX.

TABLE IX. Sampling for visual and dimensional examination AQL (approx.) - 1.5 percent defective.

Number of blades in inspection lot	Sample size, number of blades to be selected for examination (reference ASQ Z1.4)
40 and under	Inspection level III
41 to 300	Inspection level II
301 and over	Inspection level I

Note: Tightened or reduced inspection in accordance with section 7 of ASQ Z1.4 may be instituted.

4.2.3 Sampling for hardness test. A random sample of blades shall be selected from each inspection lot in accordance with the table for small sample inspection in ASQ Z1.4 at inspection level S-4 and shall be subjected to the hardness test specified in 4.4.

4.3 Visual and dimensional examination. Each of the sample blades selected in accordance with 4.2.2 shall be visually and dimensionally examined to verify compliance with this specification. Examination shall be conducted as specified in table X. Any blade in the sample containing one or more visual and dimensional defects shall not be offered for delivery. If the number of defective blades in any sample exceeds the acceptance number for the sample, this shall be cause for rejection of the lot represented by the example.

TABLE X. Classification of defects in accordance with ASQ Z1.4

Categories	Defects
Critical:	None defined.
Major:	
101	Type, class, diameter, arbor hole size, thickness, and number of teeth of saw blade not as specified
102	Blade not uniformly ground; evidence of cracks, segregations, seams, hammer marks, or high spots; blade warped
103	Blade thickness not uniform within the specified tolerance (flat ground blades only).
104	Blade thickness at rim, thinnest circle, or collar not uniform within the specified tolerance (hollow ground blades only).
105	Arbor hole diameter not within the specified tolerance.
106	Teeth not the required style.
107	Teeth not spring set and bevel filed as required; evidence of nicks, fins, or burrs.
Minor:	
201	Blade thickness not within the specified tolerance (flat ground blades only).
202	Blade thickness at rim, thinnest circle, or collar not within the specified tolerance.
203	Rim diameter not within the specified tolerance.
204	Number of teeth not within the specified tolerance.
205	Marking, manufacturer's name or trademark missing, illegible, or not permanent.

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4.3.1 Cracks and imperfection. Each of the sample blades shall be examined to determine full compliance with 3.1. Any indication of flaws, cracks, or imperfections shall be grounds for rejection. Examination shall be performed by a process subject to the approval of the bureau or activity concerned. Unless otherwise specified (see 6.2), examination shall be performed in accordance with ASTM-E 3024/E3024M.

4.4 Hardness test. Each of the sample blades selected in accordance with 4.2.3 shall be subjected to a Rockwell hardness test to determine compliance with 3.3. If any blade in the sample fails this test, this shall be cause for rejection of the lot represented by the sample.

4.4.1 Possible test failure. Possible test failures are defined as follows:

a. Hardness:

1. Hardness of blade not within the specified values on the Rockwell C scale.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that is helpful, but is not mandatory.)

6.1 Intended use. This specification covers circular saw blades for portable bench-mounted and floor-mounted woodcutting machines, and motor-driven portable tools..

6.2 Acquisition requirements. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type and class required (see 1.2).
c. Diameter of arbor hole required (see 3.5). If other than round, furnish detailed requirements.
d. Angular velocity for tensioning, if other than 10,000 s.f.m. (see 3.9).
e. Diameter, thickness, and number of teeth or teeth sections required (see applicable table).
f. Style of teeth required for type I, class 1, 30 inches and larger in diameter; and type III, class 1 (see 3.10.1 and 3.12.1).
g. When blades of the types and classes included herein are required in sizes or number of teeth not listed in the applicable tables, exceptions should be specified.
h. The number of each thickness of filler blade required in dado head (see 3.13).
i. Whether examination for cracks and imperfections shall be performed other than in accordance with ASTM-E3024/E3024M.
j. Packaging requirements (see 5.1).

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6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.4 Subject term (key word) listing.

Combination
Crosscut
Dado heads
Rip

Custodians:
Army - AV
Navy - SH
Air Force - 84

Review activities:
Other Federal - FAS

Preparing activity:
DLA - CQ

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